

Software Requirements Specification

Information System for Mart

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1. Introduction

1.1 Purpose

The purpose of the **Information System for Mart** is to develop a digital platform that helps shop owners efficiently manage products, sales, inventory, and profit tracking. The system simplifies day-to-day store operations by providing a centralized dashboard for monitoring business activities.

It aims to:

- Reduce manual record keeping
- Improve product search efficiency
- Maintain accurate stock records
- Calculate profit automatically
- Provide business insights through reports and dashboards

1.2 Scope

The system is designed for small and medium retail businesses.

The scope includes:

- Product inventory management
- Product location mapping (Rack/Shelf system)
- Sales recording
- Automatic stock updates
- Profit calculation
- Dashboard for daily business analytics
- Role-based login access for Owner and Worker

The system can be accessed through a web interface and provides real-time updates for inventory and sales.

2. Overall Description

2.1 Objectives of the Project

The main objective of the **Information System for Mart** is to create an efficient and user-friendly digital platform for managing store operations. The system is designed to simplify inventory handling, improve business monitoring, and reduce dependency on manual record keeping. It aims to enhance productivity, accuracy, and decision-making for both store owners and workers.

The specific objectives of this project are:

- **To digitize store inventory management** by replacing traditional manual records with an organized digital system for tracking products, stock levels, and item details.
- **To enable quick and efficient product searching** so users can easily locate products by name, SKU, or category, saving time during daily operations.
- **To reduce human error in stock calculations** by automatically updating inventory quantities whenever products are added, sold, or modified.
- **To monitor daily sales performance** through dashboards and reports that help owners understand sales trends, product demand, and business growth.
- **To automate profit calculation** by instantly computing profit margins based on cost price and selling price, reducing manual accounting effort.
- **To improve product organization inside the store** by using a rack and shelf mapping system, making it easier for workers to locate and manage products efficiently.
- **To provide secure access control** by implementing role-based permissions, ensuring that owners and workers have access only to the functions relevant to their responsibilities.
- **To support better business decision-making** by providing accurate records, analytics, and insights that help owners plan inventory purchases and improve overall operations.

3. Requirements

3.1 Functional Requirements

The system shall provide the following functionalities:

Product Management

- Add new products
- Update product details
- Delete products
- Categorize products
- Assign SKU codes

Search System

- Search by product name
- Search by SKU
- Search by category
- Show rack and shelf location

Sales Management

- Record product sales
- Update inventory automatically after sale
- Prevent selling more than available stock

Profit Management

- Calculate profit automatically
- Generate daily/weekly/monthly reports

User Authentication

- Secure login system
- Owner access
- Worker access

3.2 Non-Functional Requirements

Performance

- Fast search results
- Quick loading dashboard
- Real-time inventory updates

Security

- Password encryption
- Secure login authentication
- Restricted access based on roles

Reliability

- Accurate inventory records
- Consistent sales transaction handling
- Data backup support

Usability

- Simple interface
- Easy navigation
- Mobile-friendly dashboard

3.3 User Requirements

Owner

Owner can:

- Add products
- Update inventory
- View reports
- Check profits
- Manage workers

Worker

Worker can:

- Search products
- Record sales
- Check stock availability
- View product location

3.4 Software Requirements

Frontend

- React.js

Backend

- Node.js
- Express.js

Database

- PostgreSQL

Tools

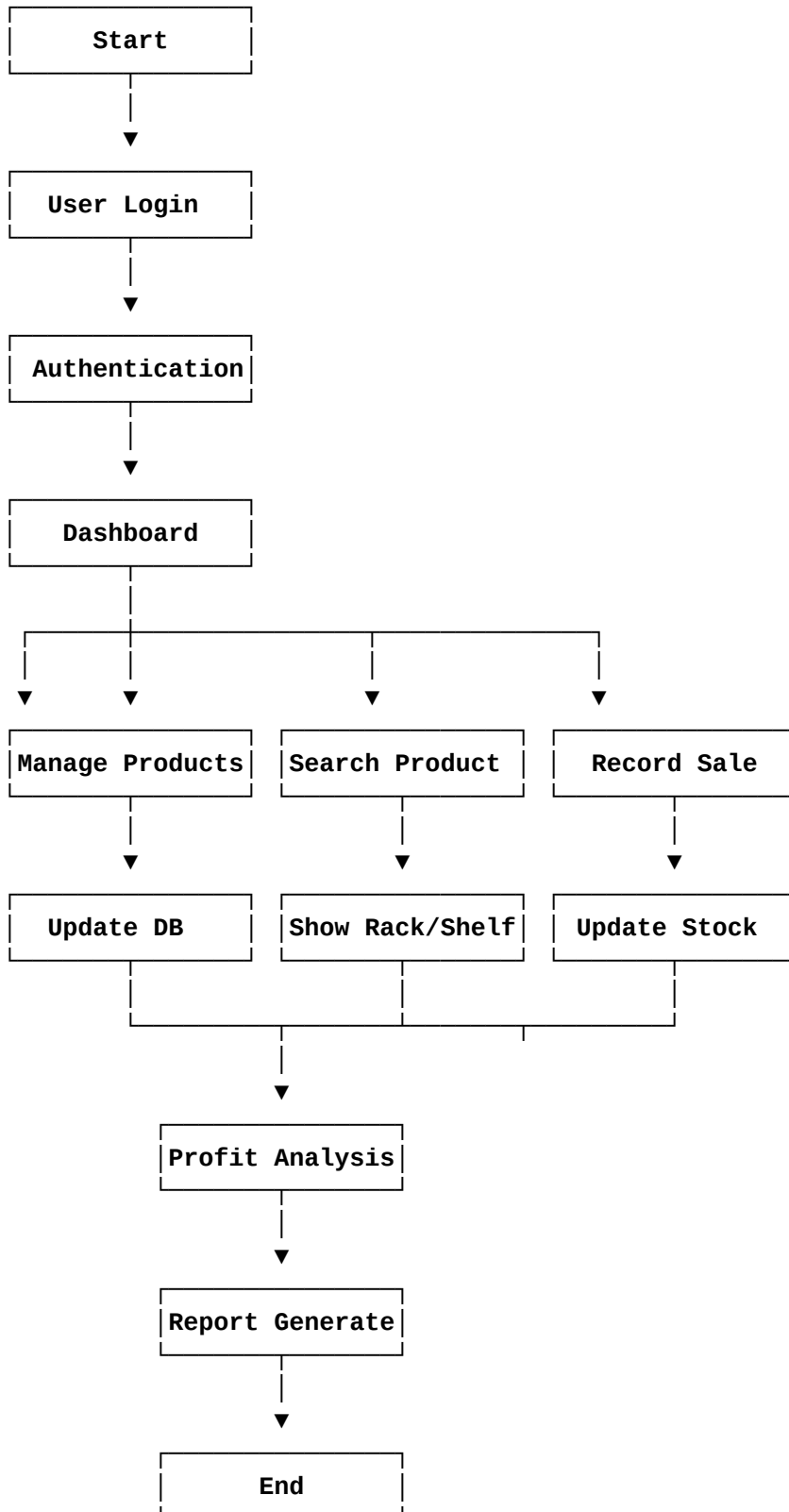
- VS Code
- Git & GitHub
- Postman

Operating System

- Windows / Linux

4. Diagrams / Flowchart

4.1 System Flowchart



5. Features

The **Information System for Mart** includes several important features designed to simplify store management, improve operational efficiency, and provide better control over daily business activities. These features help both store owners and workers perform tasks more quickly and accurately.

The main features of the system are:

- **Inventory Management**
The system allows users to add, update, and remove products from the inventory database. It keeps track of stock quantity, product details, pricing, and category information in an organized manner.
- **Smart Product Search**
Users can quickly search for products using product name, SKU code, or category. This feature helps workers locate items faster and improves customer service efficiency.
- **Rack/Shelf Product Mapping**
Each product can be assigned a physical location such as rack number and shelf position. This makes it easier for workers to find products inside the store without confusion.
- **Real-Time Stock Update**
Whenever a product is sold or updated, the inventory is automatically adjusted in real time. This ensures accurate stock records and reduces inventory mismatches.
- **Sales Recording**
The system records every sales transaction, including product details, quantity sold, date, and transaction amount, creating a complete sales history for business tracking.
- **Automatic Profit Calculation**
Profit is calculated automatically based on the difference between cost price and selling price, helping owners monitor earnings without manual calculations.
- **Daily Sales Dashboard**
A dashboard provides a quick overview of daily sales, profit, stock movement, and important business statistics for easy monitoring.
- **User Role Management**
Different access levels can be assigned to Owners and Workers, ensuring proper control over system operations and improving security.
- **Business Analytics Reporting**
The system generates useful reports on sales trends, profit margins, and product performance to support better business planning and decision-making.
- **Secure Authentication**
A secure login system protects business data by allowing only authorized users to access the platform.

6. Future Enhancement

The current system provides a strong foundation for retail business management. However, several additional features can be introduced in future versions to further improve functionality, scalability, and user experience.

Possible future enhancements include:

- **Barcode Scanner Integration**
Products can be added, searched, and sold by scanning barcodes, making operations faster and reducing manual data entry.
- **AI-Based Sales Prediction**
Artificial intelligence can analyze past sales data to predict future product demand, helping owners plan inventory more efficiently.
- **Supplier Management Module**
A dedicated supplier module can be added to manage vendor details, purchase records, supply schedules, and payment tracking.
- **Invoice Generation System**
The platform can generate automatic digital invoices or receipts for customers, improving transaction professionalism and record keeping.
- **Mobile Application Version**
A mobile app can allow users to monitor inventory, record sales, and check reports from anywhere using smartphones or tablets.
- **Low Stock Notifications**
Automatic alerts can notify owners when stock reaches a minimum threshold, helping avoid product shortages.
- **Cloud Backup System**
Cloud-based storage and backup can improve data safety, recovery, and accessibility across multiple devices.
- **Multi-Store Management Support**
The system can be expanded to handle inventory and sales management for multiple shop branches under one centralized platform.
- **Customer Purchase Analytics**
Tracking customer buying patterns can help businesses understand demand, improve product recommendations, and increase sales.
- **Advanced Reporting Dashboard**
More detailed charts, graphs, and predictive insights can provide deeper business intelligence for long-term planning.

7. Conclusion

The **Information System for Mart** provides a modern and efficient digital solution for retail business management. It is designed to simplify everyday store operations by combining inventory tracking, product searching, sales recording, and profit calculation into a single integrated platform.

By replacing traditional manual processes with an automated system, the platform helps reduce errors, save time, and improve overall productivity. Features such as real-time stock updates, organized rack and shelf mapping, secure user access, and business analytics make store management more accurate and convenient.

In addition, the system creates opportunities for future growth through advanced technologies such as barcode scanning, AI-driven forecasting, and mobile accessibility. Overall, this project serves as a practical and scalable solution that can help retail businesses operate more efficiently, make informed decisions, and achieve better long-term growth.